

Consider a two-state hidden Markov model. The states are A and B. The state transition probabilities are P_{AA} for A goes to A, P_{BB} for B goes to B, P_{AB} for A goes to B, and P_{BA} for B goes to A. When state A is unchanged, a token zero is emitted. If A changes to B, again a token zero is emitted. If state B goes to B, one is emitted, and B goes to A, one is emitted. I want to analyze this model. Let's start with the limit of rapid transitions which will be based on the notes I have attached. The question I want to address is extending that analysis to the limit where p_{AA} and P_{BB} are close to 1 so transitions are rare. In the latter limit is there a natural embedding and network?

I'm pulling the attached note into view and mapping your HMM to the long-window Gaussian framework first, then I'll turn to the rare-transition regime and what changes there.

A key simplification has appeared: in the model as you stated it, the emitted token depends only on whether the current hidden state is A or B, not on whether a transition occurred. So the observed binary process is itself just a two-state Markov chain. That makes the rare-transition limit much cleaner to analyze.

Thought for 3m 6s >

Yes. But there are really **two distinct rare-transition regimes**, and they lead to different "natural" representations.

First, let me rewrite your model in the cleanest way.

Let $z_t \in \{A, B\}$ be the hidden state, with transition matrix

$$T = \begin{pmatrix} P_{AA} & P_{AB} \\ P_{BA} & P_{BB} \end{pmatrix}, \quad P_{AA} + P_{AB} = 1, \quad P_{BA} + P_{BB} = 1.$$

Your emission rule is:

- if the current state is A , emit 0,
- if the current state is B , emit 1.

So the emitted token does **not** actually depend on whether the state changed; it depends only on which state you were in. Therefore the observed process $x_t \in \{0, 1\}$ is itself just a two-state Markov chain:

$$x_t = 0 \iff z_t = A, \quad x_t = 1 \iff z_t = B.$$

So this "HMM" is degenerate in a useful way: the hidden state is perfectly revealed by the token.

That makes the analysis much easier.

1. Basic structure of the observed process

Write

$$a := P_{AB}, \quad b := P_{BA},$$

so that

$$P_{AA} = 1 - a, \quad P_{BB} = 1 - b.$$

The stationary probabilities are

$$\pi_A = \frac{b}{a+b}, \quad \pi_B = \frac{a}{a+b}.$$

Since token 1 means state B ,

$$\mathbb{E}[x_t] = \pi_B.$$

The nontrivial eigenvalue of the chain is

$$\lambda = P_{AA} + P_{BB} - 1 = 1 - a - b.$$

This is the persistence parameter. The correlation time is of order

$$\tau_c \sim \frac{1}{1-\lambda} = \frac{1}{a+b}.$$

So:

- **rapid transitions** means $a+b$ is not small, equivalently λ not close to 1;
- **rare transitions** means $a, b \ll 1$, equivalently $\lambda \approx 1$.

For this two-state chain one has

$$\text{Cov}(x_t, x_{t+\tau}) = \pi_A \pi_B \lambda^{|\tau|}.$$

So the count of ones in a window of length L ,

$$N_1 = \sum_{t=1}^L x_t,$$

has

$$\text{Var}(N_1) = L \sigma_{\text{eff}}^2 + O(1),$$

with long-run variance

$$\sigma_{\text{eff}}^2 = \pi_A \pi_B \left(1 + 2 \sum_{\tau \geq 1} \lambda^\tau \right) = \pi_A \pi_B \frac{1 + \lambda}{1 - \lambda} = \pi_A \pi_B \frac{2 - (a + b)}{a + b}.$$

This is exactly the kind of long-run covariance object your note identifies as the source of the canonical embedding in the Gaussian window limit. In that framework, one factorizes the long-run covariance $\Sigma = EE^\top$, and the corresponding count-model is realized by an embedding lookup, sum pooling, and a quadratic head. The note also emphasizes that low-rank HMM structure gives a low-rank dependence direction and that order corrections sit beyond the bag-of-words Gaussian term.  Pasted text

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2. The Gaussian / rapid-mixing regime

If your window is long compared to the correlation time,

$$L \gg \tau_c \sim \frac{1}{a + b},$$

then the CLT picture applies well. In that case

$$\frac{N_1 - L\pi_B}{\sqrt{L}}$$

is approximately Gaussian.

Because the vocabulary has only two tokens, the count vector

$$N = (N_0, N_1), \quad N_0 + N_1 = L,$$

has only **one nontrivial direction**. So the covariance embedding is automatically one-dimensional.

A convenient centered scalar feature is

$$\phi(0) = -\pi_B, \quad \phi(1) = \pi_A,$$

since then

$$\sum_{t=1}^L \phi(x_t) = N_1 - L\pi_B.$$

The whitened feature is

$$\psi(0) = -\frac{\pi_B}{\sigma_{\text{eff}}}, \quad \psi(1) = \frac{\pi_A}{\sigma_{\text{eff}}},$$

so that

$$q = \frac{1}{\sqrt{L}} \sum_{t=1}^L \psi(x_t)$$

has asymptotic variance 1.

So in the rapid-mixing / Gaussian regime, the natural embedding is just a **scalar phase embedding**:

$$0 \mapsto -c, \quad 1 \mapsto +d,$$

with the centered-whitened choice above being the most canonical.

Then the natural network is exactly the one from your note, specialized to rank one:

embedding lookup $\rightarrow$ sum $\rightarrow$ scalar quadratic head.

Equivalently,

$$\log p(\text{window}) \approx c_0 - \frac{1}{2}q^2.$$

So in that regime the answer is very clean:

- **natural embedding**: one-dimensional centered/whitened token feature;
 - **natural network**: Deep Sets with a scalar quadratic head.  Pasted text
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3. What changes when transitions are rare

Now suppose

$$a, b \ll 1, \quad \lambda = 1 - a - b \approx 1.$$

Then the correlation time

$$\tau_c \sim \frac{1}{a + b}$$

becomes very large.

This creates a sharp distinction.

Regime A: $L(a + b) \gg 1$

If the window is still much longer than the typical run length, then the Gaussian picture survives. The only difference is that the variance is large:

$$\sigma_{\text{eff}}^2 \sim \frac{2\pi_A\pi_B}{a + b}.$$

So the same one-dimensional embedding remains natural. The bag-of-words model still works.

Regime B: $L(a + b) \lesssim 1$

This is the genuinely interesting rare-transition regime.

Now a typical window contains **zero or one switch**, not many. So the count distribution is no longer close to a single Gaussian.

In fact, if $L(a + b) \ll 1$, then with probability $1 - O(L(a + b))$, the whole window is all zeros or all ones:

- all zeros with probability approximately π_A ,
- all ones with probability approximately π_B ,

and the first correction comes from windows with one switch:

$$000 \cdots 011 \cdots 1 \quad \text{or} \quad 111 \cdots 100 \cdots 0.$$

So the count N_1 is approximately **bimodal**, with mass near 0 and L , plus a small bridge from one-switch windows.

That means:

- a **single Gaussian** on counts is not the right approximation;
- a single quadratic Deep Sets head is no longer the natural architecture.

This is exactly the kind of situation where the note's permutation-invariant Gaussian approximation stops being the right leading description, because order effects are no longer a small correction. The note explicitly distinguishes the Gaussian bag-of-words term from order-sensitive corrections.  Pasted text

4. Exact sequence model in the rare-transition regime

Because the observation reveals the state, the exact sequence likelihood is especially simple.

Let

$$N_{ij} = \#\{t = 1, \dots, L - 1 : x_t = i, x_{t+1} = j\}, \quad i, j \in \{0, 1\}.$$

Then

$$\log P(x_{1:L}) = \log \pi_{x_1} + N_{00} \log P_{AA} + N_{01} \log P_{AB} + N_{10} \log P_{BA} + N_{11} \log P_{BB}.$$

So the exact sufficient statistics are:

- the initial token x_1 ,
- the adjacent-pair counts $N_{00}, N_{01}, N_{10}, N_{11}$.

In the rare-transition limit, the most important ones are the switch counts

$$N_{01}, \quad N_{10}.$$

So the most natural representation is no longer "how many zeros and ones are in the window," but rather:

- which phase you started in,
- how long you stayed there,

- how many times you switched.

That is the metastable description.

5. Natural embedding and network in the rare-transition regime

So here is the clean answer.

If you only care about long-window bag-of-words statistics

Then the natural embedding is still **one scalar dimension**, because there are only two tokens and one nontrivial covariance direction. The network is still the Gaussian Deep Sets network from the notes:

lookup $\rightarrow$ sum $\rightarrow$ quadratic head.

But this is only natural if

$$L \gg \frac{1}{a+b}.$$

If you care about the true rare-transition regime at fixed or moderate L

Then the natural latent variable is the **phase/run state**, not just the count.

The exact minimal model is a first-order Markov chain on the observed symbols:

$$\log P(x_{1:L}) = u_{x_1} + \sum_{t=1}^{L-1} J_{x_t, x_{t+1}},$$

where

$$u_0 = \log \pi_A, \quad u_1 = \log \pi_B,$$

and

$$J = \begin{pmatrix} \log P_{AA} & \log P_{AB} \\ \log P_{BA} & \log P_{BB} \end{pmatrix}.$$

In neural-network language, that is equivalent to a tiny recurrent or pairwise model:

- either a **2-state RNN** whose hidden state stores the current phase,
- or a **bigram / pair-potential network**,
- or a **1D convolution on adjacent pairs**.

That is the natural network in the sticky regime.

The natural embedding in that regime is therefore:

- token embedding for phase: $0 \mapsto A$, $1 \mapsto B$, or numerically $0 \mapsto -1$, $1 \mapsto +1$;
- plus a transition-sensitive mechanism that scores

$$(x_t, x_{t+1}).$$

So the right answer is **not** just “an embedding,” but “an embedding plus memory of the previous token.”

6. The conceptual summary

There is a crossover controlled by

$$L(a + b).$$

If $L(a + b) \gg 1$, many switches occur inside a window. Then:

- counts are approximately Gaussian,
- the note's covariance embedding applies directly,
- the natural model is 1D embedding + sum pooling + quadratic head.  Pasted text

If $L(a + b) \ll 1$, switches are rare within a window. Then:

- counts are bimodal or mixture-like, not Gaussian,
- order is essential,
- the natural model is a sticky two-state Markov / RNN / pairwise model.

So in the rare-transition limit, there **is** a natural embedding and network, but it is different:

- **embedding**: a one-dimensional phase label is still natural;
- **network**: not Deep Sets, but a sequential model that remembers the current phase and penalizes switches.

The Gaussian embedding from your notes survives only after coarse-graining over many runs.

If you would like, I can next write this out as a compact collaborator-facing note with the exact formulas for the two regimes and the crossover at $L(a + b) \sim 1$.

Sources



Suppose I have a 2-state HMM, such that the state transition is 1-eps for self-transition and eps for other-transition. Each state has its own categorical emission probabilities over N tokens. Can I apply the transfer matrix/forward algorithm and write down analytical, closed form expression for sequence probabilities?

I'm writing it in transfer-matrix form first, then I'll separate what is exact for any sequence from what can actually be simplified in closed form.

Thought for 54s >

Yes.

For a 2-state HMM with hidden states A, B , transition matrix

$$Q = \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ \varepsilon & 1 - \varepsilon \end{pmatrix},$$

and categorical emissions

$$e_A(k) = P(x_t = k \mid z_t = A), \quad e_B(k) = P(x_t = k \mid z_t = B), \quad k = 1, \dots, N,$$

the probability of any observed sequence $x_{1:T} = (x_1, \dots, x_T)$ is exactly given by a 2×2 transfer-matrix product.

Let

$$D_k = \text{diag}(e_A(k), e_B(k)).$$

If the initial hidden-state distribution is $\pi = (\pi_A, \pi_B)^T$, then

$$P(x_{1:T}) = \mathbf{1}^T D_{x_T} Q D_{x_{T-1}} Q \cdots D_{x_2} Q D_{x_1} \pi,$$

where $\mathbf{1} = (1, 1)^T$.

That is the forward algorithm in closed matrix form.

Equivalently, defining the forward vector

$$\alpha_1 = D_{x_1} \pi, \quad \alpha_t = D_{x_t} Q \alpha_{t-1},$$

you get

$$P(x_{1:T}) = \mathbf{1}^T \alpha_T.$$

So the answer is:

- **yes**, the transfer-matrix / forward-algorithm form is exact and analytic;
- **but** for a completely arbitrary sequence $x_{1:T}$, this matrix product is usually the simplest closed form.

The reason you usually cannot simplify further is that the matrices

$$M_k := D_k Q$$

depend on the observed token k , and different M_k 's generally do **not commute**. So for a generic sequence there is no single scalar formula depending only on counts of tokens.

There are, however, useful special cases.

If the sequence is a run of the same token k repeated L times, then

$$P(k^L) = \mathbf{1}^T (D_k Q)^{L-1} D_k \pi,$$

and now you can diagonalize the single 2×2 matrix $D_k Q$. Its eigenvalues are

$$\lambda_{\pm}(k) = \frac{(1 - \varepsilon)(e_A(k) + e_B(k)) \pm \sqrt{(1 - \varepsilon)^2(e_A(k) - e_B(k))^2 + 4\varepsilon^2 e_A(k)e_B(k)}}{2}.$$

So in that case $P(k^L)$ is a linear combination of $\lambda_+(k)^{L-1}$ and $\lambda_-(k)^{L-1}$.

Also, because your transition matrix is symmetric, it helps to work in the sum/difference basis. If

$$\alpha_t = \begin{pmatrix} a_t \\ b_t \end{pmatrix}, \quad m_t = a_t + b_t, \quad d_t = a_t - b_t,$$

and for the observed symbol x_t you write

$$u_t = e_A(x_t) + e_B(x_t), \quad v_t = e_A(x_t) - e_B(x_t),$$

then with $\rho = 1 - 2\varepsilon$,

$$\begin{pmatrix} m_t \\ d_t \end{pmatrix} = \frac{1}{2} \begin{pmatrix} u_t & \rho v_t \\ v_t & \rho u_t \end{pmatrix} \begin{pmatrix} m_{t-1} \\ d_{t-1} \end{pmatrix},$$

and the sequence probability is just

$$P(x_{1:T}) = m_T.$$

So there is a very explicit 2-dimensional recursion.

The cleanest summary is:

$$P(x_{1:T}) = \sum_{z_{1:T} \in \{A,B\}^T} \pi_{z_1} \varepsilon^{N_{\text{switch}}(z)} (1 - \varepsilon)^{T-1-N_{\text{switch}}(z)} \prod_{t=1}^T e_{z_t}(x_t),$$

and the transfer-matrix formula is the efficient closed-form evaluation of that sum.

If you want, I can next derive the exact likelihood in terms of runs of identical observed symbols, which is often the most useful analytic form in the rare-transition regime.

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$$\hookrightarrow P(x_{1:T}) = \frac{1}{2} \text{TDxTQDxT} - \frac{1}{2} \text{Dx2QDx1}\pi$$

Can you help me unpack this recursive expression?

Thought for 5s >

Yes. The cleanest way to unpack it is to read the expression as "propagate a 2-dimensional belief state through time."

Let the hidden state at time t be $z_t \in \{A, B\}$, and observed token be $x_t \in \{1, \dots, N\}$.

You have

$$Q = \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ \varepsilon & 1 - \varepsilon \end{pmatrix}, \quad D_k = \begin{pmatrix} e_A(k) & 0 \\ 0 & e_B(k) \end{pmatrix},$$

where $e_A(k) = P(x_t = k \mid z_t = A)$ and $e_B(k) = P(x_t = k \mid z_t = B)$.

If the initial hidden-state distribution is

$$\pi = \begin{pmatrix} \pi_A \\ \pi_B \end{pmatrix},$$

then

$$P(x_{1:T}) = \mathbf{1}^T D_{x_T} Q D_{x_{T-1}} Q \cdots D_{x_2} Q D_{x_1} \pi, \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

What each piece means

Start with π . This is your probability of starting in A or B .

Then $D_{x_1} \pi$ means: weight each starting state by how likely it is to emit the first observed symbol x_1 .

So

$$D_{x_1} \pi = \begin{pmatrix} e_A(x_1) \pi_A \\ e_B(x_1) \pi_B \end{pmatrix}.$$

That vector is not yet the total probability; it is the pair of contributions from paths that end at hidden state A or B after explaining x_1 .

Next multiply by Q . That propagates those contributions forward one step through the hidden-state transition.

Then multiply by D_{x_2} . That weights by the probability of emitting x_2 from the new hidden state.

Repeat until time T . At the end, sum the two entries with $\mathbf{1}^T$, because the final hidden state could be either A or B .

Forward vector interpretation

Define

$$\alpha_t = \begin{pmatrix} \alpha_t(A) \\ \alpha_t(B) \end{pmatrix},$$

where

$$\alpha_t(i) = P(x_{1:t}, z_t = i).$$

Then the recursion is

$$\alpha_1 = D_{x_1} \pi,$$

and for $t \geq 2$,

$$\alpha_t = D_{x_t} Q \alpha_{t-1}.$$

Finally,

$$P(x_{1:T}) = \mathbf{1}^T \alpha_T = \alpha_T(A) + \alpha_T(B).$$

That is the forward algorithm in matrix form.

Written componentwise

For two states, this is

$$\begin{aligned}\alpha_t(A) &= e_A(x_t) [(1 - \varepsilon) \alpha_{t-1}(A) + \varepsilon \alpha_{t-1}(B)], \\ \alpha_t(B) &= e_B(x_t) [\varepsilon \alpha_{t-1}(A) + (1 - \varepsilon) \alpha_{t-1}(B)].\end{aligned}$$

And at the start,

$$\alpha_1(A) = \pi_A e_A(x_1), \quad \alpha_1(B) = \pi_B e_B(x_1).$$

So each step says:

- take previous mass in A, B ,
 - mix it according to the hidden-state transitions,
 - then multiply by the emission probability of the observed symbol.
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For $T = 2$

The formula becomes

$$P(x_1, x_2) = \mathbf{1}^T D_{x_2} Q D_{x_1} \pi.$$

Expanding:

$$\begin{aligned}P(x_1, x_2) &= e_A(x_2) [(1 - \varepsilon) \pi_A e_A(x_1) + \varepsilon \pi_B e_B(x_1)] \\ &\quad + e_B(x_2) [\varepsilon \pi_A e_A(x_1) + (1 - \varepsilon) \pi_B e_B(x_1)].\end{aligned}$$

This is exactly the sum over the four hidden paths:

$$(A \rightarrow A), (A \rightarrow B), (B \rightarrow A), (B \rightarrow B).$$

For example, the $A \rightarrow B$ path contributes

$$\pi_A \cdot e_A(x_1) \cdot \varepsilon \cdot e_B(x_2).$$

Why this equals the sum over hidden paths

The brute-force definition is

$$P(x_{1:T}) = \sum_{z_1, \dots, z_T \in \{A, B\}} P(z_1) \prod_{t=2}^T P(z_t \mid z_{t-1}) \prod_{t=1}^T P(x_t \mid z_t).$$

That has 2^T terms, one for each hidden-state path.

The matrix product is just a compact way to perform exactly that sum without writing all 2^T terms explicitly.

Each multiplication by Q sums over the previous hidden state.

Each multiplication by D_{x_t} inserts the emission factor for the observed symbol.

So the recursion compresses an exponential sum into $O(T)$ matrix-vector multiplies.

One subtle convention

There are two common conventions:

1. emit first, then transition
2. transition first, then emit

Your formula

$$D_{x_T} Q \cdots D_{x_2} Q D_{x_1} \pi$$

corresponds to:

- start in $z_1 \sim \pi$,
- emit x_1 ,
- transition,
- emit x_2 ,
- transition,
- etc.

That is perfectly standard. Just be consistent.

Intuition in words

At each time t , the 2-vector α_t stores:

- the probability of having seen the observations so far and currently being in A ,
- the probability of having seen the observations so far and currently being in B .

Then:

- Q moves probability mass between hidden states,
- D_{x_t} rewards or suppresses each hidden state according to how compatible it is with the actual observed token.

At the end, add the two possibilities.

Compact version specialized to your symmetric chain

Since

$$Q = \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ \varepsilon & 1 - \varepsilon \end{pmatrix},$$

the update is just

$$\alpha_t = \begin{pmatrix} e_A(x_t) & 0 \\ 0 & e_B(x_t) \end{pmatrix} \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ \varepsilon & 1 - \varepsilon \end{pmatrix} \alpha_{t-1}.$$

So you can think of each step as:

- first blur the previous belief by rare switching,
- then reweight by the evidence from the new symbol.

If you want, I can next rewrite this in a run-length form for the rare-transition limit $\varepsilon \ll 1$, which makes the structure even more transparent.



Does this allow for simplifications, maybe in certain limits, e.g. $\varepsilon > 0$?

I'm separating the exact algebra from the cases where it really collapses. The big lever is that your transition matrix has eigenvalues 1 and $1 - 2\varepsilon$, so one mode is conserved and the other decays.

Thought for 2m 2s >

Yes. The nicest simplification is to diagonalize the transition part.

Write

$$\alpha_t = \begin{pmatrix} \alpha_t(A) \\ \alpha_t(B) \end{pmatrix}, \quad \alpha_t(i) = P(x_{1:t}, z_t = i),$$

so

$$\alpha_t = D_{x_t} Q \alpha_{t-1},$$

with

$$Q = \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ \varepsilon & 1 - \varepsilon \end{pmatrix}, \quad D_{x_t} = \begin{pmatrix} e_A(x_t) & 0 \\ 0 & e_B(x_t) \end{pmatrix}.$$

Now define the sum and difference variables

$$m_t = \alpha_t(A) + \alpha_t(B), \quad d_t = \alpha_t(A) - \alpha_t(B).$$

Here m_t is especially nice because

$$m_t = P(x_{1:t}),$$

the prefix probability you actually care about.

Also define

$$u_t = e_A(x_t) + e_B(x_t), \quad v_t = e_A(x_t) - e_B(x_t), \quad \rho = 1 - 2\varepsilon.$$

Then the exact recursion becomes

$$\begin{aligned} m_t &= \frac{1}{2} (u_t m_{t-1} + \rho v_t d_{t-1}), \\ d_t &= \frac{1}{2} (v_t m_{t-1} + \rho u_t d_{t-1}). \end{aligned}$$

So the whole 2-state HMM reduces to a 2-scalar recursion. The parameter ε enters only through

$$\rho = 1 - 2\varepsilon.$$

That is the cleanest unpacking.

A further simplification is to divide out the total mass and track only the imbalance

$$r_t = \frac{d_t}{m_t}.$$

Then

$$\begin{aligned} m_t &= \frac{1}{2} (u_t + \rho v_t r_{t-1}) m_{t-1}, \\ r_t &= \frac{v_t + \rho u_t r_{t-1}}{u_t + \rho v_t r_{t-1}}. \end{aligned}$$

So the sequence probability is

$$P(x_{1:T}) = m_T = m_1 \prod_{t=2}^T \frac{u_t + \rho v_t r_{t-1}}{2},$$

with

$$\begin{aligned} m_1 &= \pi_A e_A(x_1) + \pi_B e_B(x_1), \\ r_1 &= \frac{\pi_A e_A(x_1) - \pi_B e_B(x_1)}{\pi_A e_A(x_1) + \pi_B e_B(x_1)}. \end{aligned}$$

That is not a closed form in raw counts, but it is an exact one-dimensional nonlinear recursion.

What simplifies in special limits:

For $\varepsilon = 0$, the hidden state never changes, so

$$P(x_{1:T}) = \pi_A \prod_{t=1}^T e_A(x_t) + \pi_B \prod_{t=1}^T e_B(x_t).$$

This is just a mixture of two i.i.d. models.

For $\varepsilon = \frac{1}{2}$, $\rho = 0$, and the recursion collapses:

$$m_t = \frac{u_t}{2} m_{t-1}.$$

Hence

$$P(x_{1:T}) = [\pi_A e_A(x_1) + \pi_B e_B(x_1)] \prod_{t=2}^T \frac{e_A(x_t) + e_B(x_t)}{2}.$$

If you also start in stationarity, $\pi_A = \pi_B = \frac{1}{2}$, then the whole observed sequence is i.i.d. with token law

$$q(k) = \frac{e_A(k) + e_B(k)}{2},$$

so

$$P(x_{1:T}) = \prod_{t=1}^T q(x_t).$$

For $\varepsilon \ll 1$, the natural simplification is an expansion in the number of hidden switches.

To first order,

$$\begin{aligned} P(x_{1:T}) &= (1 - \varepsilon)^{T-1} \left[\pi_A \prod_{t=1}^T e_A(x_t) + \pi_B \prod_{t=1}^T e_B(x_t) \right] \\ &+ \varepsilon (1 - \varepsilon)^{T-2} \sum_{s=1}^{T-1} \left[\pi_A \left(\prod_{t \leq s} e_A(x_t) \right) \left(\prod_{t > s} e_B(x_t) \right) + \pi_B \left(\prod_{t \leq s} e_B(x_t) \right) \left(\prod_{t > s} e_A(x_t) \right) \right] + O \end{aligned}$$

So the leading term is "no switch," and the first correction is "exactly one switch."

For $\varepsilon = 1$, the hidden state alternates deterministically, giving

$$P(x_{1:T}) = \pi_A \prod_{t \text{ odd}} e_A(x_t) \prod_{t \text{ even}} e_B(x_t) + \pi_B \prod_{t \text{ odd}} e_B(x_t) \prod_{t \text{ even}} e_A(x_t).$$

One more important point: for a completely arbitrary observed sequence and generic emissions, there is usually no further collapse to a formula depending only on token

counts, because the matrices $D_{x_t} Q$ for different symbols do not commute. So the exact transfer-matrix recursion is already the canonical closed form. The real simplifications come from special ε values, small- ε expansions, or special emission structure.

The most useful takeaway is:

$$\text{transition effect} \iff \rho = 1 - 2\varepsilon.$$

When $\varepsilon \approx \frac{1}{2}$, the imbalance mode is weak and the model is close to an i.i.d. mixture.

When $\varepsilon \approx 0$, the model is close to a mixture of long single-state runs.

I can next derive the small- ε likelihood in a cleaner run-based form, or write the recursion as a posterior-odds update.

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↳ I can next derive the small- ε likelihood in a cleaner run-based form, or write the recursion as a posterior-odds update.

Yes please derie the small eps limit

I'm reorganizing the likelihood by the number of hidden switches. In the rare-transition regime that is the right expansion parameter, and it makes the structure much more transparent.

Thought for 1m 8s >

Yes. In the small- ε regime, the clean way is to expand the likelihood by the **number of hidden switches**.

Let

$$a_t := e_A(x_t), \quad b_t := e_B(x_t),$$

for the observed sequence $x_{1:T}$.

Also define segment products

$$A_{u:v} := \prod_{t=u}^v a_t, \quad B_{u:v} := \prod_{t=u}^v b_t.$$

The exact likelihood is

$$P(x_{1:T}) = \sum_{z_{1:T}} \pi_{z_1} \prod_{t=2}^T Q_{z_{t-1}, z_t} \prod_{t=1}^T e_{z_t}(x_t),$$

with

$$Q = \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ \varepsilon & 1 - \varepsilon \end{pmatrix}.$$

Because every switch contributes a factor ε , and every non-switch contributes a factor $1 - \varepsilon$, you can group terms by the number K of switches:

$$P(x_{1:T}) = \sum_{K=0}^{T-1} \varepsilon^K (1 - \varepsilon)^{T-1-K} S_K(x_{1:T}),$$

where S_K is the total emission weight of all hidden paths with exactly K switches.

That is the rare-transition expansion.

First few terms

No hidden switch: $K = 0$

There are only two possibilities: always in A , or always in B . So

$$S_0 = \pi_A A_{1:T} + \pi_B B_{1:T}.$$

Hence the leading term is

$$P(x_{1:T}) = (1 - \varepsilon)^{T-1} [\pi_A A_{1:T} + \pi_B B_{1:T}] + O(\varepsilon).$$

So as $\varepsilon \rightarrow 0$, the HMM becomes just a **mixture of two i.i.d. models**.

One hidden switch: $K = 1$

If the chain starts in A and switches once after time s , the weight is

$$\pi_A A_{1:s} B_{s+1:T}.$$

If it starts in B and switches once after time s , the weight is

$$\pi_B B_{1:s} A_{s+1:T}.$$

Summing over all switch positions $s = 1, \dots, T-1$,

$$S_1 = \pi_A \sum_{s=1}^{T-1} A_{1:s} B_{s+1:T} + \pi_B \sum_{s=1}^{T-1} B_{1:s} A_{s+1:T}.$$

So keeping up to one switch gives

$$P(x_{1:T}) \approx (1 - \varepsilon)^{T-1} S_0 + \varepsilon (1 - \varepsilon)^{T-2} S_1.$$

This is usually the best "small- ε " approximation when $T\varepsilon \ll 1$.

Two hidden switches: $K = 2$

For completeness,

$$S_2 = \pi_A \sum_{1 \leq r < s \leq T-1} A_{1:r} B_{r+1:s} A_{s+1:T} + \pi_B \sum_{1 \leq r < s \leq T-1} B_{1:r} A_{r+1:s} B_{s+1:T}.$$

So the exact expansion is

$$P(x_{1:T}) = (1 - \varepsilon)^{T-1} S_0 + \varepsilon(1 - \varepsilon)^{T-2} S_1 + \varepsilon^2(1 - \varepsilon)^{T-3} S_2 + \dots$$

First-order Taylor form

If you want a literal expansion in powers of ε , expand the prefactors too:

$$(1 - \varepsilon)^{T-1} = 1 - (T - 1)\varepsilon + O(\varepsilon^2),$$

so

$$P(x_{1:T}) = S_0 + \varepsilon[S_1 - (T - 1)S_0] + O(\varepsilon^2).$$

That is the fixed- T , $\varepsilon \rightarrow 0$ asymptotic series.

Why this is the natural rare-transition picture

In the sticky regime, the hidden path is mostly one long run, with occasional switch points.

So the likelihood is dominated by:

- **zero-switch paths**,
- then **one-switch paths**,
- then **two-switch paths**, etc.

That is much more informative than the Gaussian / bag-of-words picture.

A useful computational simplification

Define prefix products

$$A_s := A_{1:s} = \prod_{t=1}^s a_t, \quad B_s := B_{1:s} = \prod_{t=1}^s b_t.$$

Then

$$A_{1:s} B_{s+1:T} = A_s \frac{B_T}{B_s}, \quad B_{1:s} A_{s+1:T} = B_s \frac{A_T}{A_s},$$

so

$$S_1 = \pi_A B_T \sum_{s=1}^{T-1} \frac{A_s}{B_s} + \pi_B A_T \sum_{s=1}^{T-1} \frac{B_s}{A_s},$$

whenever those ratios are well-defined.

In log form, define

$$\ell_s := \sum_{t=1}^s \log \frac{a_t}{b_t}.$$

Then

$$\frac{A_s}{B_s} = e^{\ell_s},$$

and

$$S_1 = \pi_A e^{\sum_{t=1}^T \log b_t} \sum_{s=1}^{T-1} e^{\ell_s} + \pi_B e^{\sum_{t=1}^T \log a_t} \sum_{s=1}^{T-1} e^{-\ell_s}.$$

This makes the one-switch term very interpretable: it is controlled by the cumulative evidence favoring A versus B .

Interpretation

The dominant zero-switch term is

$$\pi_A \prod_{t=1}^T e_A(x_t) + \pi_B \prod_{t=1}^T e_B(x_t),$$

which says:

- either the whole sequence was generated in state A ,
- or the whole sequence was generated in state B .

The one-switch correction says:

- maybe the sequence starts looking like A ,
- then at some hidden time s , it starts looking like B ,
- or vice versa.

So in the rare-transition limit the sequence is naturally described by:

- a starting state,
 - a small number of switch points,
 - long emission segments between switches.
-

When this approximation is valid

The expected number of hidden switches in $T - 1$ steps is about

$$(T - 1)\varepsilon.$$

So:

- if $(T - 1)\varepsilon \ll 1$, keeping only S_0 or $S_0 + S_1$ is very good;
- if $T\varepsilon = O(1)$, you need several S_K terms;
- if $T\varepsilon \gg 1$, this rare-switch expansion is no longer the right simplification.

That last case is where the Gaussian / long-window description starts to re-emerge.

The next natural step is deriving the small- ε **log-likelihood** and the approximate **MAP switch time** from the $S_0 + S_1$ truncation.



OK. Then I would like to proceed to some intermediate regime, where we can derive an effective interaction between the switchings (domain walls).

I'm mapping the posterior over hidden states to a 1D Ising chain first. That makes "switchings = domain walls" exact, and then the effective wall interaction drops out cleanly.

Thought for 3m 31s >

Yes. The clean way to do it is to look at the **posterior over hidden paths given the observed sequence**. That posterior is exactly a 1D Ising chain in a random field, and the switchings are domain walls.

One important caveat first:

If you average over all observations, there is **no extra interaction** between switches beyond the Markov prior. The nontrivial interaction appears only **conditioned on a realized sequence** $x_{1:T}$.

Let me derive it.

Take hidden state

$$s_t \in \{+1, -1\}, \quad +1 \equiv A, \quad -1 \equiv B.$$

The transition matrix is

$$P(s_{t+1} = s_t) = 1 - \varepsilon, \quad P(s_{t+1} = -s_t) = \varepsilon.$$

Let the emission probabilities be

$$e_+(x) = e_A(x), \quad e_-(x) = e_B(x).$$

Then for a fixed observation sequence $x_{1:T}$,

$$P(s_{1:T} \mid x_{1:T}) \propto \pi_{s_1} \prod_{t=1}^{T-1} Q_{s_t, s_{t+1}} \prod_{t=1}^T e_{s_t}(x_t).$$

Now write the transition weight as

$$Q_{s, s'} = \sqrt{\varepsilon(1-\varepsilon)} \exp(J s s'), \quad J = \frac{1}{2} \log \frac{1-\varepsilon}{\varepsilon}.$$

And write the emission as

$$e_{s_t}(x_t) = \sqrt{e_A(x_t)e_B(x_t)} \exp(h_t s_t),$$

with

$$h_t = \frac{1}{2} \log \frac{e_A(x_t)}{e_B(x_t)}.$$

So the posterior becomes

$$P(s_{1:T} \mid x_{1:T}) \propto \exp \left[J \sum_{t=1}^{T-1} s_t s_{t+1} + \sum_{t=1}^T h_t s_t + h_0 s_1 \right],$$

where

$$h_0 = \frac{1}{2} \log \frac{\pi_A}{\pi_B}.$$

So this is exactly a **1D random-field Ising model**.

Now define the domain-wall variables

$$n_t = \frac{1 - s_t s_{t+1}}{2} \in \{0, 1\}.$$

Then $n_t = 1$ means there is a switch between t and $t + 1$.

Since

$$s_t s_{t+1} = 1 - 2n_t,$$

the coupling term becomes

$$J \sum_{t=1}^{T-1} s_t s_{t+1} = J(T-1) - 2J \sum_{t=1}^{T-1} n_t.$$

So each wall has a bare cost

$$2J = \log \frac{1 - \varepsilon}{\varepsilon}.$$

At the prior level, walls are just a dilute gas with fugacity

$$y = e^{-2J} = \frac{\varepsilon}{1 - \varepsilon}.$$

The interesting part comes from the field term. If you reconstruct the spin from the wall history,

$$s_t = s_1 \prod_{u < t} (1 - 2n_u),$$

then

$$\sum_{t=1}^T h_t s_t = s_1 \sum_{t=1}^T h_t \prod_{u < t} (1 - 2n_u).$$

That is the exact wall action. It is nonlocal in the wall variables.

To see the effective interaction cleanly, order the walls:

$$\tau_1 < \tau_2 < \cdots < \tau_K.$$

Suppose the background phase is A on the left. Then a B -domain is an interval bounded by two neighboring walls,

$$(\tau_{2j-1}, \tau_{2j}].$$

Take one such interval $(u, v]$. Compare:

- path with no walls on that interval, so it stays in A ,
- path with two walls, so it flips to B on $(u, v]$ and then back to A .

The exact Bayes factor is

$$\frac{P(B\text{-domain on } (u, v] \mid x)}{P(\text{no domain on } (u, v] \mid x)} = \left(\frac{\varepsilon}{1 - \varepsilon} \right)^2 \prod_{t=u+1}^v \frac{e_B(x_t)}{e_A(x_t)}.$$

So the corresponding effective pair potential between the two walls is

$$U_A(u, v \mid x) = 2 \log \frac{1 - \varepsilon}{\varepsilon} + \sum_{t=u+1}^v \log \frac{e_A(x_t)}{e_B(x_t)}.$$

Equivalently, in terms of h_t ,

$$U_A(u, v \mid x) = 2\mu + 2 \sum_{t=u+1}^v h_t, \quad \mu := \log \frac{1 - \varepsilon}{\varepsilon}.$$

Likewise, for an A -domain inside a B -background,

$$U_B(u, v \mid x) = 2\mu - 2 \sum_{t=u+1}^v h_t.$$

That is the effective interaction you were asking for.

So the picture is:

- each wall costs μ ,
- two adjacent walls enclosing an opposite-phase domain interact through the cumulative log-likelihood ratio across the interval between them.

For multiple domains, the weight factorizes over disjoint intervals, plus the obvious hard-core ordering constraint:

$$u_1 < v_1 < u_2 < v_2 < \dots.$$

So in the interval gas language, the posterior is a gas of disjoint opposite-phase domains with weights

$$w_A(u, v \mid x) = e^{-U_A(u, v \mid x)}, \quad w_B(u, v \mid x) = e^{-U_B(u, v \mid x)}.$$

Now the “intermediate regime” interpretation.

Under data actually generated from state A ,

$$\mathbb{E}_A \left[\log \frac{e_A(X)}{e_B(X)} \right] = D_{\text{KL}}(e_A \| e_B).$$

Therefore

$$\mathbb{E}_A[U_A(u, v \mid x)] = 2\mu + (v - u) D_{\text{KL}}(e_A \| e_B).$$

Similarly under B ,

$$\mathbb{E}_B[U_B(u, v \mid x)] = 2\mu + (v - u) D_{\text{KL}}(e_B \| e_A).$$

So on average the two walls feel a **linear confining potential**:

$$U(\ell) \sim 2\mu + \gamma\ell,$$

with string tension

$$\gamma_A = D_{\text{KL}}(e_A \| e_B), \quad \gamma_B = D_{\text{KL}}(e_B \| e_A).$$

This is the key result.

The walls want to annihilate because separating them creates a domain whose emission law disagrees with the data, and the disagreement accumulates linearly with length.

There is also disorder around that mean.

Define

$$\Lambda_t := \log \frac{e_A(x_t)}{e_B(x_t)}.$$

Then for an A -background,

$$U_A(u, v \mid x) = 2\mu + \sum_{t=u+1}^v \Lambda_t.$$

For a long interval of length $\ell = v - u$,

$$\sum_{t=u+1}^v \Lambda_t \approx \ell D_{\text{KL}}(e_A \| e_B) + \sqrt{\ell \Delta_A} \eta,$$

where

$$\Delta_A = \text{Var}_{X \sim e_A} \left[\log \frac{e_A(X)}{e_B(X)} \right].$$

So in the intermediate regime the effective pair potential is

$$U_A(\ell) \approx 2\mu + \gamma_A \ell + \sqrt{\Delta_A \ell} \eta.$$

That is: linear confinement plus random fluctuations.

This gives a natural crossover length:

$$\ell_* \sim \frac{2\mu}{\gamma} = \frac{2 \log((1 - \varepsilon)/\varepsilon)}{D_{\text{KL}}(e_\sigma \| e_{-\sigma})}.$$

Interpretation:

- for $\ell \ll \ell_*$, the bare wall cost dominates;
- for $\ell \gg \ell_*$, the emission evidence dominates and long wrong-state domains are strongly suppressed;
- around $\ell \sim \ell_*$, you are in the intermediate regime where the wall gas is dilute but definitely interacting.

So the clean answer is:

- the exact posterior over hidden states is a random-field Ising chain;
- switchings are domain walls;
- the effective interaction between two neighboring walls at $u < v$ is the interval potential

$$U_{\sigma}(u, v \mid x) = 2 \log \frac{1 - \varepsilon}{\varepsilon} + \sum_{t=u+1}^v \log \frac{e_{\sigma}(x_t)}{e_{-\sigma}(x_t)};$$

- on average this becomes a linear confining interaction with slope given by a KL divergence.

That is the natural domain-wall theory for the intermediate regime.

The next natural step is to write the full sequence likelihood as an exact partition sum over disjoint domains and then take either a continuum limit or a saddle-point approximation for the typical wall spacing.



↳ the exact posterior over hidden states is a random-field Ising chain;

This is nice result, but I was hoping to have the marginal pdf over the emitted sequences. Or do you think from the stats of the states one can compute the emission and get the token probabilities easily? Also the h_t field can only take two values depending on which token gets emitted, right?

Thought for 1m 8s >

Yes — the clean separation is:

- the **domain-wall / Ising picture** is most natural for the **posterior over hidden states** $P(s_{1:T} \mid x_{1:T})$,
- the **marginal law of the emitted sequence** $P(x_{1:T})$ is the corresponding **partition function** after summing over all hidden states.

For your 2-state HMM with

$$Q = \begin{pmatrix} 1 - \varepsilon & \varepsilon \\ \varepsilon & 1 - \varepsilon \end{pmatrix},$$

and emissions $e_A(k), e_B(k)$ over $k = 1, \dots, N$, define spins

$$s_t = +1 \text{ for } A, \quad s_t = -1 \text{ for } B.$$

Then for a fixed observed sequence $x_{1:T}$,

$$P(x_{1:T}, s_{1:T}) = \pi_{s_1} \prod_{t=1}^{T-1} Q_{s_t, s_{t+1}} \prod_{t=1}^T e_{s_t}(x_t).$$

Now write

$$J = \frac{1}{2} \log \frac{1 - \varepsilon}{\varepsilon}, \quad h(k) = \frac{1}{2} \log \frac{e_A(k)}{e_B(k)}, \quad h_0 = \frac{1}{2} \log \frac{\pi_A}{\pi_B}.$$

Also

$$\begin{aligned} Q_{s,s'} &= \sqrt{\varepsilon(1 - \varepsilon)} e^{Jss'}, \\ e_s(k) &= \sqrt{e_A(k)e_B(k)} e^{h(k)s}, \\ \pi_s &= \sqrt{\pi_A\pi_B} e^{h_0s}. \end{aligned}$$

So the **exact marginal sequence probability** is

$$P(x_{1:T}) = \left[\sqrt{\pi_A\pi_B} (\varepsilon(1 - \varepsilon))^{\frac{T-1}{2}} \prod_{t=1}^T \sqrt{e_A(x_t)e_B(x_t)} \right] Z(x_{1:T}),$$

where

$$Z(x_{1:T}) = \sum_{s_{1:T} \in \{\pm 1\}^T} \exp \left[J \sum_{t=1}^{T-1} s_t s_{t+1} + \sum_{t=1}^T h(x_t) s_t + h_0 s_1 \right].$$

That is the exact marginal pdf over emitted sequences.

So yes: the emitted-sequence law is an Ising **partition function in a token-dependent field**.

Now to your two questions.

First: can you “compute the emission and get token probabilities easily from the stats of the states”?

For **low-order token marginals**, yes.

If the hidden chain is in stationarity, here it is symmetric so

$$\pi_A = \pi_B = \frac{1}{2}.$$

Then the one-token marginal is just

$$P(x_t = k) = \frac{1}{2} e_A(k) + \frac{1}{2} e_B(k).$$

The two-token marginal at lag τ is

$$P(x_t = k, x_{t+\tau} = l) = \sum_{i,j \in \{A,B\}} \pi_i(Q^\tau)_{ij} e_i(k) e_j(l).$$

Since for this symmetric chain

$$Q^\tau = \frac{1}{2} \begin{pmatrix} 1 + \rho^\tau & 1 - \rho^\tau \\ 1 - \rho^\tau & 1 + \rho^\tau \end{pmatrix}, \quad \rho = 1 - 2\varepsilon,$$

this becomes

$$P(x_t = k, x_{t+\tau} = l) = p(k)p(l) + \frac{\rho^\tau}{4} \Delta(k)\Delta(l),$$

with

$$p(k) = \frac{e_A(k) + e_B(k)}{2}, \quad \Delta(k) = e_A(k) - e_B(k).$$

So token correlations are easy: they inherit the same decay factor $(1 - 2\varepsilon)^\tau$ as the hidden chain.

But for the **full sequence probability** $P(x_{1:T})$, low-order state statistics are not enough. You need the full transfer-matrix / partition-function expression above. In general the emitted process is itself **not Markov of order 1**.

Second: does h_t only take two values?

In general, **no**.

You have

$$h_t = h(x_t) = \frac{1}{2} \log \frac{e_A(x_t)}{e_B(x_t)}.$$

So each token k carries its own field value

$$h(k) = \frac{1}{2} \log \frac{e_A(k)}{e_B(k)}.$$

Therefore:

- for an alphabet of size N , h_t can take **up to N distinct values**;
- if two tokens have the same likelihood ratio $e_A(k)/e_B(k)$, they give the same field;
- only in the **binary-token case** does h_t take two values;
- if some emission probability is zero, then $h(k) = \pm\infty$, which is a hard constraint rather than a finite field.

So the random field is not "two-valued" unless your emission alphabet is binary.

The useful conceptual picture is:

- the hidden coupling J comes from switching cost,
- the token x_t injects a local field $h(x_t)$,
- the marginal emitted-sequence probability is the partition function of that 1D random-field Ising chain.

If you want the next step, the most useful thing is to derive the exact transfer matrix in the Ising variables and then take an intermediate-regime approximation for $P(x_{1:T})$ in terms of rare but interacting domains.

